um-game-playing-strength-of-mcts-variants

# [7th Solution] Ensemble of Tree + NN

| | |
|---|---|
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| **Version** | Original |

Source: https://www.kaggle.com/competitions/um-game-playing-strength-of-mcts-variants/discussion/549617
Retrieved with Kaggle CLI on 2026-07-03.

- CV
  Well since for me sometimes GKF with Game more algin on cv LB and sometimes GKF with GamerulesetName better. I am a bit confused and in the final days I stick with GKF + Game + statify with label, for I think for test set **new Game dominate**, but this might be a wrong decsion as PB show my final ensemble weights using this CV not work at all.
- Data aug
  Like other teams I have also used flip aug (swich agent1 agent2, advantageP1,label) at very first and it show improve on LB about 0.01
  I also used test time TTA by combining flipped data.
- Tree models
  For me tree model work better then nn and tree model results is much more stable on LB.
  The key for tree models to perform well is using

1. **manual cross feats of numerical feats** like * / of top 20 numercial feats
   This improve both cv and LB a lot, it keep improve using top 5 , 10 15 until 20, then more cross like top 25 * 25 will hurt LB.

```
                                                             ('ap1/100/
    INFO     tree-main.py:408 in main()
             df_fi.head(10):                             feat_name     importance
                         975                     agent1_selection  131665.505185
                         974                          AdvantageP1  128173.017510
                         973                     agent2_selection  127067.497154
                         972         AdvantageP1(/)Completion   82469.000426
                         971         AdvantageP1(*)Completion   73456.791266
                         970  OutcomeUniformity(*)Completion   41925.539571
                         969                           Asymmetric   39836.317985
                         968                       agent2_playout   39471.525783
                         967            agent2_exploration_const   38419.259947
                         966                       agent1_playout   36777.249548
```

2. Target encode help LB 0.001
3. Make tree deeper for lgb I used 16 and for cbt I used 10.
4. To avoid overfit set reg lambda high like 10, set random strength of cbt high like 10.
5. Dart mode for lgb improve cv but hurt LB, my cv GKF + Game might not correct here. Well trust LB…
6. Tree results:
   LGB single model 10 folds LB 419 PB 425
   CBT single model 5 folds LB 422 PB 428 I tried so much effort in the last week to improve CBT model however it not contribute to final PB.
   Well simple ensemble of LGB + CBT improve LB only a bit less then 0.001 but not improve PB.

- NN models
  Follow the https://www.kaggle.com/code/yekenot/mcts-deeptables-nn this great notebook.
  I used torch to impement the NN model with top numeric feats turn to bins and adding lgb tree leafs feats(depth 5, 200 trees).
  Also used CIN + FM for the pooling layer and used label aug(mutinomal) for nn model.
  However nn model not stable for LB and work worse on CV (GKF + Game)

By luck I have one 5 folds NN model with LB 419 PB 427, notice due to randomness nn model mostly get LB about 423, but since LB and PB align you just use your best LB nn model.

- Post deal
  Optimize oof a * x + b for overall rmse
- Ensemble
  My best LB 416 PB 423 model is simple blend of LGB(10 folds) + NN (5 folds) with weights 1, 0.75.

If I have more time, I will try to swich back to cv with GKF + Gamerulesetname and test CV + LB and try stacking with OOF method.

## Supplemental Q&A

> **Andreas Bisiadis · 2024-12-03T05:58:38.453000**
>
> @goldenlock What do you mean by
>
> > Optimize oof a * x + b for overall rmse
>
> Why does it help more than the original predicted target?

> **gezi · 2024-12-03T06:19:52.323000**
>
> help shift pred distribution you could see mean pred trun to simliar mean of label

> **Vasilis · 2024-12-03T10:01:06.677000**
>
> Congrats on solo gold! what about this: "manual cross feats of numerical feats like * / of top 20 numercial feats
> This improve both cv and LB a lot, it keep improve using top 5 , 10 15 until 20, then more cross like top 25 * 25 will hurt LB.". Does this mean that you get the top 10 most important feature and multiply or divide them with each other?

> **gezi · 2024-12-03T10:22:55.140000**
>
> yes exactly.

> **stefanoclss · 2024-12-03T12:40:58.857000**
>
> Maybe one little question for these / * in top numerical feats, did you check for correlation?

> **gezi · 2024-12-03T20:11:04.887000**
>
> No I just juse top 20 features from a depth 6 lgb model, here the candidats really matters for example similar depth 6 cbt model output top 20 features work much worse, still has room to improve.
> Also here is why for me I found lgb work better then cbt, after adding 20 * 20 cross feats lgb work better on LB.

> **Vasilis · 2024-12-05T17:42:21.110000**
>
> Hi @goldenlock , another question, what do you mean by target encoding?

> **gezi · 2024-12-06T04:39:13.623000**

> @vasileioscharatsidis For each agent specific category feature(each cat value) calc it's mean utility_agent1 as feature, notice this does not improve my cv but help 0.001 on LB as cv does not drop much(nearly the same) I accepted this change.